ARC MSP: Adaptations for Anthropic

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In [1] the authors designed an estimator which estimates the expected output of a wide random MLP with Gaussian (He) initialisation. More specifically, they proved the following theorem.

Theorem 1 ([1, Theorems 5.1 and 5.2])**.** *Fix an integer $L \geq 0$ and polynomial activation functions $\phi_1, \dots, \phi_L$. Given weights $\theta \in \mathbb{R}^{(L+1)\times n\times n}$ drawn independently from $\mathcal{N}\big(0, \frac{2}{n}\big)$ for an integer $n \geq 1$, write $M_\theta \colon \mathbb{R}^n \to \mathbb{R}^n$ for the MLP with L hidden layers, width n, activation functions $\phi_1, \dots, \phi_L$ and weights θ.*[1] *Then there is an estimation procedure $G(\theta, \varepsilon)$ such that for every noticeable*[2] *function $0 < \varepsilon(n) \leq 1$ and every output index i, as $n \to \infty$,*

$$\mathbb{E}_\theta\Big[\big(G(\theta, \varepsilon(n))_i - \mathbb{E}_{X\sim\mathcal{N}(0,I_n)}[M_\theta(X)]_i\big)^2\Big] = O\big(\varepsilon(n)^2\big)$$

and $G(\theta, \varepsilon)$ runs in time $O(n^2/\varepsilon^2)$. If we also require that $\varepsilon(n)^2 = O(1/n^2)$, then $G(\theta, \varepsilon)$ can be made to run in time $O(n/\varepsilon^2)$.

For comparison, Monte Carlo sampling with $\lceil 1/\varepsilon^2 \rceil$ samples achieves the same mean squared error in time $\Theta(n^2/\varepsilon^2)$ [1, Section 2.3]. This shows that at initialisation their estimator can outperform random sampling. A question they leave open [1, Section 7.2] is whether their methods can be adapted to outperform sampling throughout training. A common heuristic, supported in idealised settings and by experiments [2, 3], is that the weights at the end of training are distributed approximately according to a (localised) Bayesian posterior. It is not exact, since SGD's anisotropic gradient noise changes its stationary distribution [4]. This distribution on the weights is used in singular learning theory [5, 6] to probe the local loss landscape geometry. The problem we propose is whether the methods from [1] can be used to prove that Theorem 1 holds when the weights are distributed according to the Gibbs posterior of a fixed dataset, with a Gaussian prior, which is the exact distribution SLT uses for probing the local geometry.

Question 1. *Fix an integer $L \geq 0$ and polynomial activation functions $\phi_1, \dots, \phi_L$, and for each $n \geq 1$ fix a dataset $\mathcal{D} = \{(x_j, y_j)\}_{j=1}^m \subset \mathbb{R}^n \times \mathbb{R}^n$, a loss function $\ell \colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ that is bounded below, a centre $\theta_0 \in \mathbb{R}^{(L+1)\times n\times n}$, a localisation strength $\gamma > 0$ and an inverse temperature $\beta \geq 0$. Let p_0 denote the density of the Gaussian $\mathcal{N}(\theta_0, \gamma^{-1}I)$ on $\mathbb{R}^{(L+1)\times n\times n}$, so that $p_0(\theta) \propto \exp\big(-\frac{\gamma}{2}\|\theta-\theta_0\|^2\big)$, and let*

$$\pi_{\beta,\mathcal{D}}(\theta) \propto \exp\Big(-\beta\sum_{j=1}^m \ell\big(M_\theta(x_j), y_j\big)\Big)\, p_0(\theta)$$

be the corresponding Gibbs posterior. Is there an estimation procedure $G_{\beta,\mathcal{D}}(\theta, \varepsilon)$, which may also depend on ℓ, θ_0 and γ, such that for every noticeable function $0 < \varepsilon(n) \leq 1$ with $\varepsilon(n)^2 = O(1/n^2)$

[1] That is, $M_\theta(x) = W^{(L+1)}\phi_L\big(W^{(L)}\cdots\phi_2\big(W^{(2)}\phi_1\big(W^{(1)}x\big)\big)\cdots\big)$ for $\theta = \big(W^{(1)}, \dots, W^{(L+1)}\big)$, with the activation functions applied coordinatewise and no bias terms [1, Section 2.1].

[2] That is, $\varepsilon(n) \geq n^{-k}$ for some $k \geq 0$ and all sufficiently large n.

and every output index i, as $n \to \infty$,

$$\mathbb{E}_{\theta\sim\pi_{\beta,\mathcal{D}}}\Big[\big(G_{\beta,\mathcal{D}}(\theta,\varepsilon(n))_i - \mathbb{E}_{X\sim\mathcal{N}(0,I_n)}[M_\theta(X)]_i\big)^2\Big] = O\big(\varepsilon(n)^2\big),$$

and which runs in time $o\big(n^2/\varepsilon(n)^2\big)$, i.e. faster than the $\Theta(n^2/\varepsilon^2)$ Monte Carlo baseline? In particular, is this achieved by the estimation procedure G of Theorem 1 itself, used unchanged with $\theta \sim \pi_{\beta,\mathcal{D}}$? For which centres θ_0, losses ℓ and regimes of m, β and γ as functions of n is this possible?

Remark 1 (Progress on Question 1). The unpublished note [7] answers Question 1 in the affirmative for all outputs outside the loss, with the estimators of [1] used unchanged, in the following case: $\theta_0 = 0$ and $\gamma = n/2$, so that p_0 is the density of the weights in Theorem 1; a fixed $\beta > 0$; the squared loss $\ell(z,y) = \sum_{o\in S}(z_o - y_o)^2$ on a set S of output coordinates of fixed size; inputs and targets with $\|x_j\|^2 \le C_x n$ and $|(y_j)_o| \le y_{\max}$ for fixed C_x and $y_{\max}$; and $n \ge m^\kappa$ for a fixed $\kappa > 2 + \max\{1,(q-1)L\}$, where q is the maximal degree of $\phi_1,\dots,\phi_L$. Precisely [7, Theorem C and Corollary C.1], the estimator of [1] of cumulant order K (its polynomial or basic Hermite algorithm), whose mean squared error under p_0 is $O(n^{-K})$, has mean squared error $O(n^{-K})$ under $\pi_{\beta,\mathcal{D}}$ for every output $i \notin S$. For $i \in S$ its mean squared error is $O(m\,n^{-K})$, and the estimator of cumulant order $K+1$ brings this back to $O(n^{-K})$. All implied constants are independent of n, m and the data. □

In [8], under a series of approximations, the author proves that at any point in training the weights of an MLP are distributed according to a Gaussian whose mean and covariance change according to

$$\begin{aligned} \theta &\sim \mathcal{N}\big(\bar\theta(t), \Sigma(t,\tau)\big), \\ \frac{d}{dt}\bar\theta(t) &= -\nabla\mathcal{L}\big(\bar\theta(t)\big), \qquad \Sigma(t,\tau) = \int_0^\tau e^{-H(t)s}\,D(t)\,e^{-H(t)s}\,ds, \end{aligned} \tag{1}$$

see [8, Definition 2.1 and eq. (67)]. Here θ is viewed as a vector in $\mathbb{R}^{(L+1)n^2}$. The flow is started at the left end $a := t - \Delta/2$ of a short window $[t-\Delta/2, t+\Delta/2]$ around t, from the weights W_a of the SGD trajectory at that time, $\bar\theta(a) = W_a$, and the law (1) is conditional on W_a. Moreover, for the dataset $\mathcal{D}$ and the loss ℓ of Question 1, now assumed twice continuously differentiable,

$$\ell_j(\theta) = \ell\big(M_\theta(x_j), y_j\big), \qquad \mathcal{L} = \frac{1}{m}\sum_{j=1}^m \ell_j, \qquad \xi_j = \nabla\ell_j - \nabla\mathcal{L}, \qquad C^{\mathrm{grad}} = \frac{1}{m}\sum_{j=1}^m \xi_j\xi_j^\top,$$

$$H(t) = \nabla^2\mathcal{L}\big(\bar\theta(t)\big), \qquad D(t) = \frac{\eta}{b}\,C^{\mathrm{grad}}\big(\bar\theta(t)\big),$$

where η is the learning rate and b the batch size of SGD, each batch consisting of b independent uniform draws from $\mathcal{D}$. The trajectory time t corresponds to t/η steps of SGD, and $\tau > 0$ is a duration of local time over which the SGD noise accumulates around the point $\bar\theta(t)$, with the Hessian $H(t)$ and the diffusion $D(t)$ frozen at their values there. Thus the mean of the law is the noise-free gradient flow of the training loss, restarted from the actual weights at the beginning of the window, and its covariance is the SGD noise accumulated over local time τ and shaped by the local Hessian.

Can the methods of [1] be adapted to prove a version of Theorem 1 where the weights are distributed according to (1)?

Question 2. *Fix an integer $L \ge 0$, polynomial activation functions $\phi_1,\dots,\phi_L$, a dataset $\mathcal{D}$ and a twice continuously differentiable loss function ℓ as in Question 1, a learning rate $\eta > 0$, a batch*

size $b \geq 1$, a trajectory time $t > 0$, a window length $\Delta \in (0, 2t]$, weights $W_a \in \mathbb{R}^{(L+1)\times n\times n}$ at the start $a = t - \Delta/2$ of the window, and a local duration $\tau > 0$, and let $\mu_{t,\tau} := \mathcal{N}\big(\bar{\theta}(t), \Sigma(t,\tau)\big)$ be the law (1) given W_a. Is there an estimation procedure $G_{t,\tau}(\theta, \varepsilon)$, which may depend on $\mathcal{D}$, ℓ, η, b, t, Δ, W_a and τ, such that for every noticeable function $0 < \varepsilon(n) \leq 1$ with $\varepsilon(n)^2 = O(1/n^2)$ and every output index i, as $n \to \infty$,

$$\mathbb{E}_{\theta\sim\mu_{t,\tau}}\Big[\big(G_{t,\tau}(\theta,\varepsilon(n))_i - \mathbb{E}_{X\sim\mathcal{N}(0,I_n)}[M_\theta(X)]_i\big)^2\Big] = O\big(\varepsilon(n)^2\big),$$

and which runs in time $o\big(n^2/\varepsilon(n)^2\big)$, i.e. faster than the $\Theta(n^2/\varepsilon^2)$ Monte Carlo baseline? In particular, is this achieved by the estimation procedure G of Theorem 1 itself, used unchanged with $\theta \sim \mu_{t,\tau}$? For which weights W_a, losses ℓ and regimes of Δ, τ, m and η/b as functions of n is this possible?

References

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